

Mark M. Mammetkuli

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QUALIFICATIONS SUMMARY: Physical chemist with 9 years of experience in modelling (C, Matlab, LabView), solid state lasers, dye lasers, optical alignment, optical detectors (CCD, PMT), lab equipments (oscilloscope, function generator, Lock-in Amplifier, Integrators, power meters), quantum chemistry, vibrational spectroscopy, UV-Vis, FTIR, Raman and 3 years of experience in vacuum systems, cryostats.

Experience

PRODUCT DEVELOPMENT – Applied and won SBIR Phase I grant (200K) from Department of Energy in 2021. As a principal investigator in this 9-month project, I performed experiments to generate proof-of-concept experimental data for Raman spectroscopy probe which can be used in turbid solutions. I supervised a team of 3 people. Application for Phase II funding is being prepared and will be submitted in April 2022. As Phase I work funded by NASA, performed preliminary experimental work for new interferometry being developed at Intelligent Optical Systems INC. Based on the results of our preliminary work, NASA awarded Phase II award for 750K for 2 years to further develop the technology.

LASERS - Aligned and maintained 10 ns 1064 nm Nd:YAG laser and cw 527 nm Nd:LIF laser. Replaced flashlamp, electronics board in Nd:YAG laser. Pumped dye laser with Nd:YAG to produce 10 ns pulses in visible region and studied electronic levels of boron containing compounds. Replaced LBO crystal in Nd:LIF laser since antireflective coating had burnt spots. Periodically replaced water and filters in chillers. Used He:Ne laser for alignment.

DETECTORS – Used PMT with monochromators to detect light at specific wavelength. Analyzed data with gated integrator to add time resolution. Calibrated monochromator with Hg arc lamp. Wrote drivers in C language that digitized signals from Lock-in amplifier and stored in computer.

OPTICAL ALIGNMENT - Aligned Pockels cell in Nd:YAG using Maltese cross method. Set up optical oscillator using Ti:sapphire nonlinear crystal to be pumped with Nd:LIF solid state laser at 527 nm cw beam and generate fs pulses centered at 798 nm. Fed weak fs pulses from optical oscillator into aligned amplifier setup which included pulse stretcher, amplifier and pulse compressor. Used optical parametric amplification (OPA) and difference frequency generator (DFG) to generate fs IR pulses. Periodically aligned optical setup that included polarizers, steering mirrors, parabolic mirrors, lenses, notch filters to perform Sum Frequency Generation spectroscopy of air/water interface.

MODELLING/DATA ANALYSIS – Wrote code in C to study experimental setup of vibrating wire viscometer. Specifically, analyzed wire length, thickness, current flow and Magnetic field on the experimental outcome. Based on this analysis, I built a vibrating wire viscometer in CNRS Grenoble France and used it to study viscosity of liquid helium at high pressures. During my graduate study at USC, I wrote a program in Matlab to study experimental geometry of our laser. Specifically, I found ideal incoming laser angles for visible and IR lasers to be used in our experiments. This work will be published soon. I wrote another code in Matlab to extract molecular orientation of water at the air/water interface. Final draft of this work is written and will be submitted to a high impact journal. I wrote a driver for CCD in Labview that will collect the data, analyze it and store it in a local folder. Modelled Ti:sapphire oscillator with OSLO to increase conversion efficiency.

LAB EQUIPMENTS – Typical equipments that I used in optical experiments are PMT, CCD, Lock-in Amplifier, Gated integrator, Function generator, power meters, Autocorrelators, oscilloscope, voltage/current/resistance meters.

PUBLIC SPEAKING – Presented my work on molecular orientation of water at air/water interface at Pacific Conference of Spectroscopy and Dynamics (PCSD 2020) in San Diego. Collaborated with theory group led by Francesco Paesani at UCSD. Presented progresses made in my research to a group of professor and students every 4 month. Volunteered twice as a judge in science competition organized by Magnolia Science Academy.

Work History

06/20-Present **Physical Chemist**, Intelligent Optical Systems, INC, Torrance, CA, USA
09/14-05/20 **Researcher**, Department of Chemistry, University of Southern California, CA, USA.
05/15-08/15 **Internship**, Gilead Sciences, Inc, San Dimas, CA, USA.
10/12-07/13 **Internship**, G2ELab, CNRS, Grenoble, France.
09/11-05/12 **Research Assistant**, IRIS, California State University, Northridge, CA, USA.
05/11-08/11 **Internship**, NHK Laboratories, Santa Fe Springs, CA, USA.

Education

09/14-05/20 **PhD in Physical Chemistry (GPA 3.3)**, University of Southern California, LA, CA, USA.
09/11-12/13 **MS Degree in Chemistry (GPA 3.8)**, California State University, Northridge, CA, USA.
09/08-12/10 **BS Degree in Chemistry (GPA 2.8)**, California State University, Northridge, CA, USA.

Skills

Languages	Python, C, Matlab, Mathematica, Gaussian, Gamess, LabView, Latex, Machine Learning
Platforms	Linux Fedora, Ubuntu, Windows
Instruments	Continuum ND/YAG Solid State Laser, Dye Laser, Spectra Physics Solid State Laser, Ti/Sapphire Oscillator, CCD, Optical Parametric Amplifier, NDFG, Vacuum Pumps, Lock-in Amplifier, PMT, Boxcar/Gated Integrator, Raman spectrometer, UV-Vis, GCMS, FTIR

Publications

- 2021 **Nature, Third Author**, Field-Dependent Orientation of D2O at an Electrode Surface Observed via SFG Spectroscopy”, *Submitted*
- 2021 **Nature, Third Author**, Asymmetric response of interfacial water to positive vs. negative electric fields.
- 2019 **Journal of Chemical Physics, Second Author**, Re-Orientation of Water Molecules in Response to Surface Charge at Surfactant Interfaces.
- 2019 **Journal of Physical Chemistry Letters, Third Author**, Molecular Orientation of Poly-3-Hexylthiophene at the Buried Interface with Fullerene.
- 2017 **Journal of Physical Chemistry B, Second Author**, Insight into Water Structure at the Surfactant Surfaces and in Microemulsion Confinement.
- 2013 **The Journal of Chemical Physics, Second Author**, Dynamics of vortex assisted metal condensation in superfluid helium.
- 2013 **Master’s Thesis**, Viscosity measurements of helium-4 using vibrating wire as a viscometer.
- 2010 **Journal of Environmental Science and Technology, Second Author**, Kinetics of UV-H₂O₂ Advanced Oxidation in the Presence of Alcohols: The Role of Carbon Centered Radicals.